CMPE 48A Term Project – Phase 1 Report

Cloud-Native Deployment of *Local History Story Map* on Google Cloud Platform

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1. Team

Role	Name
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2. Application Code

Repository: https://github.com/AysuKeskin/storymap-cloud-native

The application is *Local History Story Map*, a platform where users discover and publish location-based community stories through a map, a feed, a timeline and full-text search. We reuse the application code that was developed earlier in the CMPE 354 software-engineering course (original upstream: https://github.com/bounswe/bounswe2026group4). The cloud-native design, the Kubernetes manifests, Terraform code, Cloud Functions and Locust tests are new work for this course.

Layer	Technology
Backend API	Django 4.2 + Django REST Framework, Gunicorn, JWT auth
Web frontend	React + Vite, served by nginx
Mobile client	Expo / React Native (consumes the same API)
Database	MySQL 8.0
Already containerised	Dockerfiles for backend and nginx/frontend, Docker Compose

Main API domains: users and authentication, stories (map / feed / timeline / search), media uploads, comments and likes, tags, notifications, gamification (badges) and a geocoding proxy to OpenStreetMap Nominatim.

Changes planned to make the application cloud-native

1. Move media storage from the local disk to **Cloud Storage** with `django-storages`, so that API pods are stateless and can be scaled freely.
2. Replace the per-process in-memory cache with a shared **Redis** cache. This also gives a distributed rate-limit lock for the Nominatim proxy, which today works only within one process.
3. Move e-mail sending out of the request path. The API publishes an event to **Pub/Sub** and a **Cloud Function** sends the e-mail.
4. Add **image thumbnail generation** as a Cloud Function triggered by uploads to the bucket.
5. Read all configuration from Kubernetes Secrets and ConfigMaps, and run database migrations as a Kubernetes `Job`.

3. Tentative Cloud Architecture

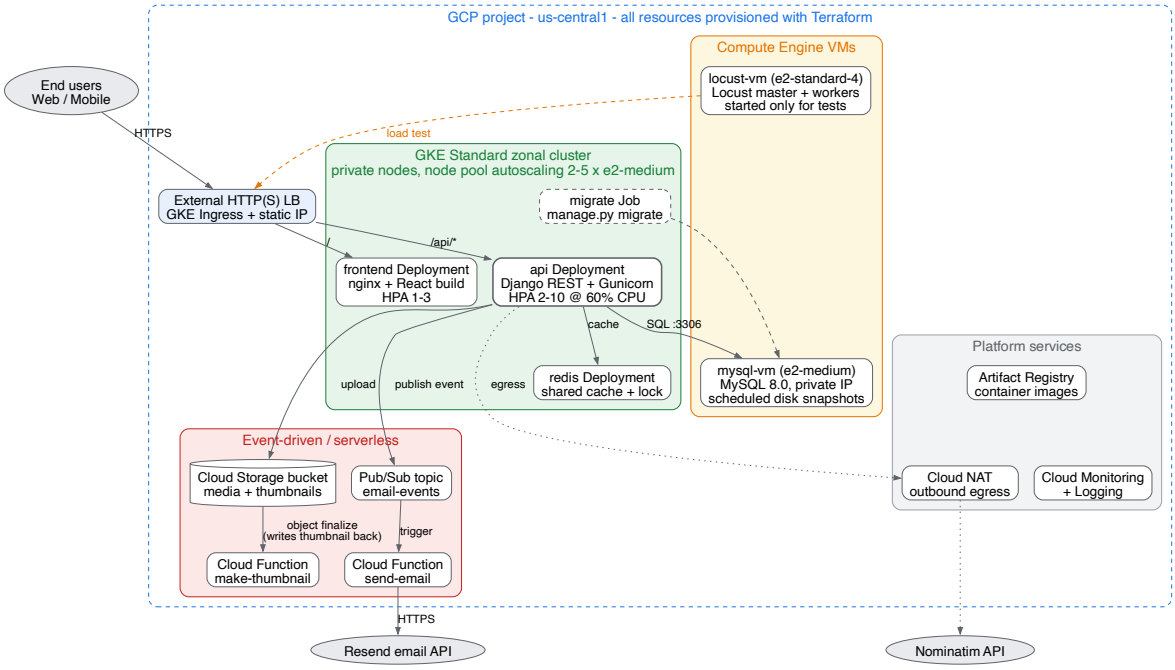


Figure 1: Tentative GCP architecture. All resources are provisioned with Terraform.

3.1 Components

Component	GCP service	Role in the system
Load balancer	External HTTP(S) LB (created by GKE Ingress), static IP	Single public entry point. Routes `/api/*` to the API service and `/` to the frontend.
frontend	GKE Deployment (+ HPA 1–3)	nginx serving the compiled React app. Lightweight and stateless.
api	GKE Deployment (+ HPA 2–10 at 60 % CPU)	Django REST API on Gunicorn. Main scaling target and the focus of the performance tests.
redis	GKE Deployment + ClusterIP Service	Shared cache for hot read endpoints (`/stories/map`, `/stories/feed`, `/stories/timeline`), geocoding results and a distributed lock. Runs in-cluster instead of Memorystore to save budget.
migrate	Kubernetes Job	Runs `manage.py migrate` before each rollout.
mysql-vm	Compute Engine VM (e2-medium)	MySQL 8.0 on a private IP, reachable only from inside the VPC. Persistent disk with a scheduled snapshot policy. Chosen over Cloud SQL for cost and to satisfy the VM requirement with a stateful, functional role.
locust-vm	Compute Engine VM (e2-standard-4)	Runs Locust master and workers. Started only during experiments so that load is generated outside the cluster it is measuring.
Media bucket	Cloud Storage	Stores user-uploaded images, audio and video, plus generated thumbnails.

Component	GCP service	Role in the system
make-thumbnail	Cloud Function (2nd gen), GCS *object finalize* trigger	Creates resized thumbnails for every uploaded image and writes them back to the bucket.
email-events	Pub/Sub topic	Asynchronous queue that decouples the API from e-mail delivery (registration verification, password reset, notification e-mails).
send-email	Cloud Function (2nd gen), Pub/Sub trigger	Consumes `email-events` and sends the message through the Resend e-mail API. Retries on failure.
Artifact Registry	Docker repository	Stores the `api` and `frontend` images built by GitHub Actions.
Cloud NAT	Cloud Router + NAT	Outbound internet access for private GKE nodes and VMs (Nominatim, package installs). Nothing in the VPC has a public IP.
Monitoring / Logging	Cloud Monitoring, Cloud Logging	CPU, memory, pod count, LB latency and error metrics used in the performance evaluation.

3.2 Interactions

1. **Read path.** A request goes Client → LB → `api` pod. The pod checks Redis first. On a cache miss it queries MySQL on the VM and stores the result in Redis with a short TTL.
2. **Write path (story with media).** The `api` pod writes the story row to MySQL and streams the file to Cloud Storage. The bucket's *finalize* event triggers `make-thumbnail`, which writes the thumbnail next to the original.
3. **Asynchronous e-mail.** Registration and notification flows publish a small JSON message to `email-events` and return immediately. `send-email` then delivers the message. A slow e-mail provider no longer slows down API requests.
4. **Scaling.** The HPA adds `api` pods when average CPU exceeds 60 %. The GKE cluster autoscaler adds nodes (2 → 5) when pods cannot be scheduled.
5. **Deployment.** Terraform creates the VPC, NAT, GKE cluster, VMs, bucket, Pub/Sub topic, Cloud Functions, Artifact Registry and a billing budget. Kubernetes manifests (Deployments, Services, HPA, Ingress, Job) are then applied with `kubectl`.

3.3 Planned performance experiments (preview)

Locust user classes will simulate (a) anonymous visitors browsing the map, feed and story details, (b) logged-in users who like and comment, and (c) content creators who post stories with images.

Independent variables	Dependent variables
Number of concurrent users and spawn rate	p50 / p95 / p99 latency
Read vs. write traffic mix	Throughput (requests per second)
Redis cache on vs. off	Error rate
HPA on (2–10 pods) vs. fixed replicas	CPU and memory of pods and the MySQL VM
Synchronous vs. Pub/Sub e-mail	Pod and node count over time

4. Cost Estimation

Prices are **list prices for `us-central1`** taken from the GCP pricing pages. They will be re-checked in the GCP Pricing Calculator before deployment. We assume about **two months** of usage (mid-November to mid-January).

4.1 Monthly baseline (system running 24/7)

Resource	Configuration	Unit price	Monthly cost
GKE management fee	1 zonal Standard cluster	\$0.10/h (≈ \$73)	**\$0.00** (covered by the \$74.40/month GKE free-tier credit)
GKE nodes	2 × e2-medium (2 vCPU, 4 GB)	\$0.0335/h each	\$48.92
Node boot disks	2 × 30 GB pd-standard	\$0.04/GB-month	\$2.40
mysql-vm	1 × e2-medium	\$0.0335/h	\$24.46
mysql-vm disk + snapshots	20 GB pd-balanced + ∼5 GB snapshots	\$0.10 and \$0.05/GB-month	\$2.25
HTTP(S) Load Balancer	1 forwarding rule	\$0.025/h	\$18.25
Cloud NAT	3 VM endpoints + low traffic	\$0.0014/h per VM + \$0.045/GB	\$3.57
Cloud Storage	∼2 GB Standard	\$0.020/GB-month	\$0.04
Artifact Registry	∼2 GB images (0.5 GB free)	\$0.10/GB-month	\$0.15
Cloud Functions (2nd gen)	< 100k invocations/month	free tier: 2M req., 180k vCPU-s	\$0.00
Pub/Sub	< 1 GB/month	first 10 GiB free	\$0.00
Monitoring and Logging	within free allotment (50 GiB logs)	–	\$0.00
Internet egress	∼5–10 GB	∼\$0.12/GB	∼\$1.00
Total per month			**≈ \$101**

4.2 Load-testing campaign (one-off, about 30 test hours)

Resource	Usage	Cost
locust-vm (e2-standard-4)	30 h × \$0.134/h + 20 GB disk	\$4.82
Extra GKE nodes from autoscaling	up to 3 × e2-medium × 30 h	\$3.02
LB data processing and egress	∼50 GB × \$0.008/GB + misc.	\$1.50
Cloud Functions / Pub/Sub under load	still within the free tier	\$0.00
Total		**≈ \$9.50**

4.3 Project total vs. the $300 credit

Scenario	Calculation	Total	Share of $300
Worst case: always on	2 × $101 + $9.50	≈ **$212**	71 %
Planned: shut down when idle	baseline ∼50 % of the time + tests	≈ **$110–120**	∼40 %

Cost-control measures

- `terraform destroy`, or scaling the node pool to 0 and stopping `mysql-vm` when the system is not needed. Everything can be recreated from code in about 15 minutes.
- `locust-vm` runs only during experiments and can use **Spot** pricing (∼60–90 % cheaper).
- Small e2 machine types and 30 GB pd-standard boot disks instead of the 100 GB default.
- Redis and MySQL are self-hosted, which avoids Memorystore (∼$35+/month) and Cloud SQL (∼$25–50/month).
- A **billing budget with alerts at 50 %, 75 % and 90 %** of $300 is created by Terraform (`google_billing_budget`).